

## Lecture 12

### Modern Telecommunication Systems and Measurements

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## Topics Covered (Chapter 15–22)

- Communication Hardware
- Internet Technologies
- Satellite Communication
- Telecommunication Systems
- Optical Communication
- Cellular Communication
- Wireless Technologies
- Communication Measurements and Tests

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## Communication Hardware

- Transmission Line
  - Wire, cable, coaxial cables
- Antenna
  - Dipole, monopole, loop, microstrip, helix, aperture, etc.
- Microwave Components
  - Waveguide, Resonators, Diodes, Tubes
- Other hardware
  - EMI/EMC, shielding, etc.

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## Internet Technologies

- Application of Internet
  - Email, File, WWW, E-commerce, Image, VoIP, Video, etc.
  - Telemedicine, video-conferencing
- Cloud Computing
  - Cloud-based design and manufacturing
- Internet of Things
  - Machine-to-Machine Communication
- Internet Security

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## Satellite Communication

- Satellite orbits
- Satellites
- Ground Stations
- Satellite Applications
  - Global Positioning System,

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## Telecommunication Systems

- Telephone
- Facimile
- Paging Systems
- Internet Telephony (VoIP)

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## Optical Communication

- Total Internal Reflection
- Fibre Cable
- Laser, LED, wavelength-division multiplexing
- Passive Optical Network (PON)
- Power Budget
- Photonic Integrated Circuit, All Optical Communication

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## Cellular Communication

- First generation cellular network (1G)
  - AMPS
- Second generation cellular network (2G)
  - GSM, IS-95
- Third generation cellular network (3G)
  - WCDMA, CDMA-2000, TD-SCDMA,
- Fourth generation cellular network (4G)
  - LTE, LTE-A, 5G...

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## Wireless Technologies

- Wireless LAN
- PANs and Bluetooth
- ZigBee and Mesh Wireless Networks
- WiMAX and Metropolitan-Area Networks
- Infrared Wireless
- Radio-Frequency Identification and Near-Field Communications
- Ultrawideband Wireless

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## 22-1: Communication Test Equipment

- There are many different types of test instruments available for use with communication equipment.
- Conventional test equipment such as multimeters, signal generators, and oscilloscopes are used along with specialized communication instruments.
- In communication work, technicians continue to use standard oscilloscopes and multimeters for measuring voltages, currents, and resistance.

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## 22-1: Communication Test Equipment

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- Voltage Measurements
  - RF Voltmeters, RF Probes, Oscilloscopes,
- Power Measurements
  - Dummy load, SWR meter, signal generator, function generator, sweep generator, RF function generator, frequency counter, spectrum analyser, network analyser, field strength meter,
- Transmitter Test
  - Frequency test, modulation test, power measurement, antenna and transmission line test
- Receiver Test
  - Noise test, power output test, sensitivity test, blocking, third-order intercept test, etc.
- Other Tests

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## Vista

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*It is dangerous to put limits on wireless.*

– Guglielmo Marconi, 1932.

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